Hanyeol Ryu

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RESEARCH INTERESTS

- Low-power wireless communications (e.g. backscatter communications)
- Channel modeling & estimation
- AI/ML for intelligent wireless communications
- Wireless security & privacy
- Software-defined radio

EDUCATION

Pusan National University
- *B.S. in Electronics Engineering* — Feb 2026

RESEARCH EXPERIENCE

Pusan National University
- *Undergraduate Researcher (PI: Prof. Sangkil Kim)* — Jan 2024 – Feb 2026
- *Research Assistant (PI: Prof. Sangkil Kim)* — Mar 2026 – Present

PROFESSIONAL ACTIVITIES

- TPC Reviewer, ***IEEE International Symposium on Intelligent Signal Processing and Communication Systems (ISPACS)***, 2026.

PUBLICATIONS

Published Articles (†: First author, *: Corresponding author)
- **J4: H. Ryu†**, N. Ha, and S. Kim*, "Transformer-Hypernetwork-Controlled Deep-Unfolded Phase-Aware Channel Estimation Refinement for Phase-Drift-Robust Backscatter Links," *arXiv preprint,* Jun. 2026. (DOI: 10.48850/arXiv.2606.31400).
- **J3:** N. Ha†, **H. Ryu**, H. Jeong, H. Kim, G. Kim, A. Georgiadis, M. M. Tentzeris, and S. Kim*, "Integrated Sensing and Backscatter Communication (ISABC): Toward Self-Sustainable Autonomous IoT Networks," *IEEE Microwave Magazine,* Accepted, 2026.
- **J2**: **H. Ryu†**, and S. Kim*, "Ultra-low-power Monostatic Backscatter Platform with Phase-Aware Channel Estimation and System-Level Validation," *IEEE Internet of Things Journal*, May 2026. (DOI: 10.1109/JIOT.2026.3696951)
- **J1**: J. Choi†, **H. Ryu†**, and S. Kim*, "Improved Backscattering Communication Range Using Reflection Amplifiers," *The Journal of Korean Institute of Electromagnetic Engineering and Science,* Feb. 2025. (DOI: 10.5515/KJKIEES.2025.36.3.266)
- **C2**: **H. Ryu†**, and S. Kim*, "Deep Learning-Aided Phase-Aware Channel Estimation for Error-Floor Suppression over Phase-Drifting Backscatter Links," *IEEE International Symposium on Intelligent Signal Processing and Communication Systems (ISPACS)*, Accepted, 2026.
- **C1**: **H. Ryu†**, and S. Kim*, "High-Isolation Yagi-Uda Antenna Array using Split-Ring Resonators for Practical Backscatter System," *IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting (AP-S/URSI)*, 2025. (DOI: 10.1109/AP-S/CNC-USNC-URSI55537.2025.11266818)

PATENTS

- Korean Patent: *"Apparatus and Method for Phase-Aware Channel Estimation in Monostatic Backscatter Communication Systems"*, Apr. 2026.

CONFERENCE PRESENTATIONS

Title: Deep Learning-Aided Phase-Aware Channel Estimation for Error-Floor Suppression over Phase-Drifting Backscatter Links (Oral)
- *IEEE ISPACS* Oct 2026

Title: Ultra-low-power Backscatter Platform with Phase-Aware Channel Estimation (Oral)
- *The Korean Institute of Electromagnetic Engineering and Science (KIEES)* Feb 2026

Title: High-Isolation Yagi-Uda Antenna Array using Split-Ring Resonators for Practical Backscatter System (Oral)
- *IEEE AP-S/URSI* Jul 2025

Title: Improved Backscattering Communication Range Using Reflection Amplifiers (Poster)
- *KIEES* Feb 2025

HONORS & AWARDS

- 2^{nd} Place: Hackathon, **Pusan National University**, Aug. 2023.
- 2^{nd} Place: Capstone Design Project, **Pusan National University**, Nov. 2025.

SERVICE & EXTRACURRICULAR ACTIVITIES

University

Pusan National University
- *Treasurer,* ***Student Council*** Dec 2021 – Dec 2022
- *Student Ambassador,* ***Pusan National University*** Apr 2021 – Feb 2023
- *Volunteer,* ***Overseas Volunteering Program (3D Printer & Arduino) — Thailand*** Dec 2022 – Feb 2023

Military Service

Republic of Korea Marine Corps (ROKMC)
- *Squad Leader,* ***ROKMC Special Reconnaissance*** Feb 2017 – Jul 2018
- 2^{nd} Place, ***ROKMC Special Reconnaissance Selection & Qualification Training*** Jun 2017 – Aug 2017

SKILLS & TECHNIQUES

- **Programming:** MATLAB, Python, C++
- **AI/ML:** PyTorch, TensorFlow/Keras
- **RF/EM Simulation:** Ansys HFSS, Advanced Design System (ADS)
- **SDR Platform:** GNU Radio
- **Hardware Design:** OrCAD PCB Editor, OrCAD Capture
- **3D Modeling:** Rhino 7